AFRICAN INSTITUTE FOR MATHEMATICAL SCIENCES (AIMS Rwanda)
Course: Model Fitting and Validation **Group:** Group 3
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1 Introduction

Mathematical models based on systems of ordinary differential equations (ODEs) are used to describe biological and epidemiological processes. Reliable parameter estimation is crucial for making predictions with such models.

The aim of this project is to estimate the parameters of Aron's density malaria model using synthetic data and to quantify uncertainty using Bayesian inference through Markov chain Monte Carlo (MCMC). Classical least squares and Bayesian approaches were compared, parameter identifiability was assessed through a local uncertainty analysis, and posterior uncertainty was propagated to produce predictive distributions.. All simulations and fitting were performed in MATLAB [2].

2 Model Description and Synthetic Data Generation

Aron's density malaria model describes within host malaria dynamics as a function of host age α. The model consists of three state variables: parasite density $p(\alpha)$, an immune response proxy $r(\alpha)$, and gametocyte density $g(\alpha)$. The equations are:

$$\frac{dp}{d\alpha} = v - (r + r_b)p, \quad \frac{dr}{d\alpha} = 0.2p - \beta r,$$
$$\frac{dg}{d\alpha} = 0.5p \exp\big(-(r + r_b)\big) - \delta g.$$

The unknown parameter vector is $\theta = (\theta_1, \theta_2, \theta_3, \theta_4) = (v, r_b, \beta, \delta)$, where v represents the parasite production rate, r_b is the baseline immune clearance rate, β is the immune decay rate, and δ is the gametocyte clearance rate. The model is solved over the age interval $\alpha \in [0, 25]$ with initial conditions $p(0) = 1, \quad r(0) = 0, \quad g(0) = 0$. Synthetic observations were generated by solving the model using the given true parameter values, $\theta = (6, 0.66, 0.005, 0.4)$, and adding independent Gaussian noise with standard deviation $\sigma = 0.05$ to the model outputs for all three state variables. The observation model is

$$y(\alpha_i) = f(\alpha_i; \theta) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

The synthetic data retain the trends in each component: $p(\alpha)$ decays with age, $r(\alpha)$ increases steadily, and $g(\alpha)$ peaks early and then declines toward zero as seen in Figure 1.

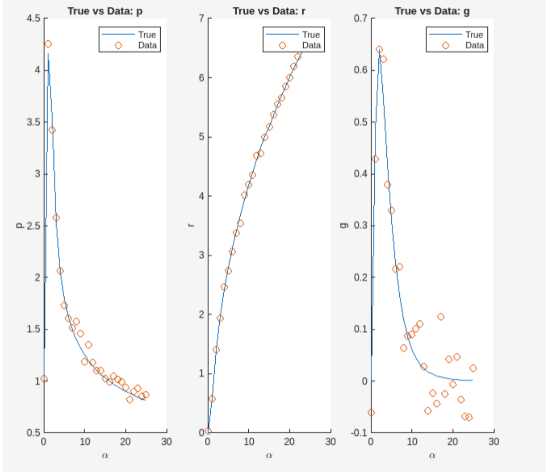


Figure 1: True model trajectories and noisy synthetic observations.

3 Least Squares Parameter Estimation and Local Uncertainty Analysis

The unknown parameters were estimated by minimizing the least squares objective function using all three outputs (p, r, g). The least squares estimates were $\hat{\theta} = (\hat{v}, \hat{r}_b, \hat{\beta}, \hat{\delta}) = (6.2778,\ 0.7553,\ 0.0062,\ 0.3551)$. The residual sum of squares was $SS(\hat{\theta}) = 0.24789$, giving $\hat{\sigma}^2 = 0.0033498$ (so $\hat{\sigma} \approx 0.058$), which is close to the simulation noise level.

Figure 2 shows that the fitted trajectories match the noisy observations closely for $p(\alpha)$ and $r(\alpha)$. For $g(\alpha)$, the model captures the early peak and decay. A local uncertainty analysis was performed using a numerical approximation of the Jacobian matrix of the model outputs with respect to the parameters. The estimated covariance matrix was

$$\mathrm{Cov}(\hat{\theta}) = \begin{pmatrix} 0.0098 & 0.0032 & 0.0000 & -0.0010 \\ 0.0032 & 0.0014 & 0.0000 & -0.0004 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ -0.0010 & -0.0004 & 0.0000 & 0.0006 \end{pmatrix}.$$

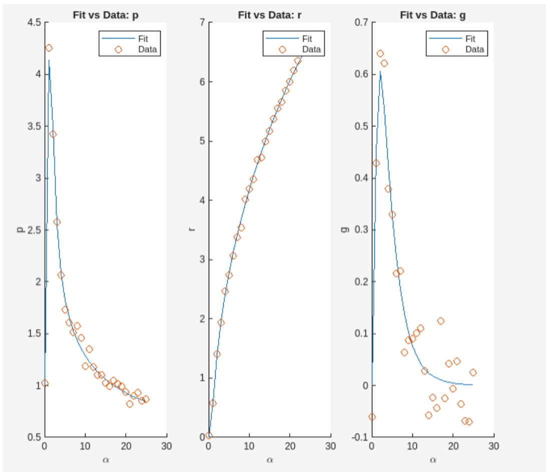


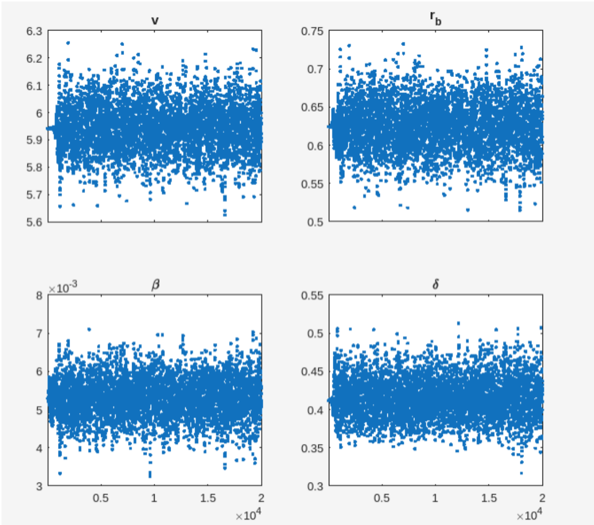
Figure 2: Least squares fitted trajectories compared to noisy data.

The corresponding standard deviations and t−values are shown in Table 1. All t−values are large (greater than 3), indicating that the parameters are identifiable under the chosen noise level.

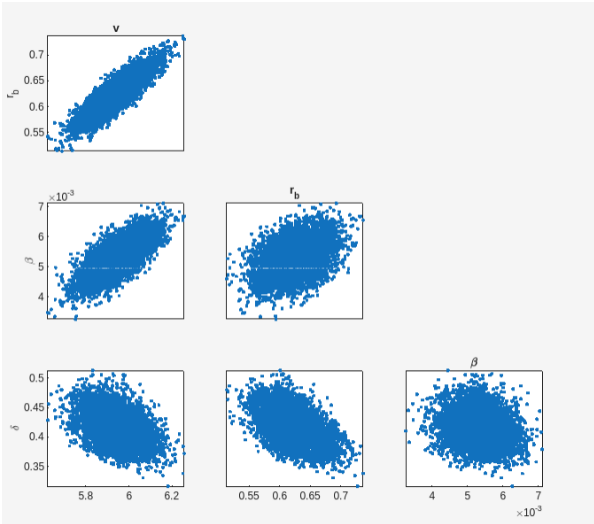
Table 1: Least squares estimates with Jacobian based uncertainty.

Parameter	Estimate	Std. Dev.	t–value
v	6.2778	0.0988	63.5208
r_b	0.7553	0.0368	20.5502
β	0.0062	0.0006	10.8164
δ	0.3551	0.0253	14.0324

4 Bayesian Inference Using MCMC



(a) Trace plots of the MCMC samples for each parameter.



(b) Posterior pair plot showing dependence between each of the parameter.

Figure 3: MCMC diagnostics for Aron's malaria density model.

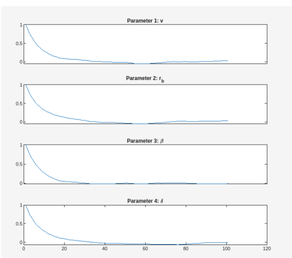


Figure 4: Auto correlation plot

Bayesian parameter estimation was performed using an adaptive Metropolis MCMC algorithm, implemented with the mcmcstat toolbox [1]. A total of 20 000 samples were generated. The trace plots in figure 3(a) show stable sampling behaviour without visible drift, consistent with good mixing. Figure 3(b) shows moderate correlations between some parameters, a positive association between $\theta_1 = v$ and $\theta_2 = r_b$, indicating compensatory effects in the model. However, the distributions remain compact and approximately elliptical, suggesting that all parameters are practically identifiable. The rejection rate was approximately 69%, corresponding to an acceptance rate of about 31%, which is reasonable. Also, Figure 4 shows that the parameter are identifiable because it shows convergence.

5 Posterior Results and Predictive Uncertainty

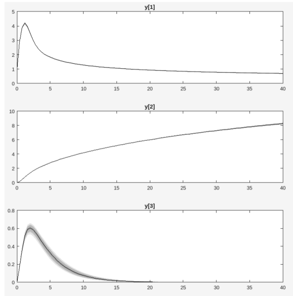


Figure 5: Posterior predictive simulations for $p(\alpha)$, $r(\alpha)$, and $g(\alpha)$.

Posterior means and 95% credible intervals were computed from the MCMC samples. All true parameter values lie within the corresponding credible intervals as shown in Table 2, indicating successful recovery of the parameters. Posterior samples were propagated through the model to generate predictive distributions. The resulting trajectories illustrate increasing uncertainty outside the calibration as shown in Figure (5) where y[1], y[2] and y[3] are $p(\alpha)$, $r(\alpha)$, and $g(\alpha)$ respectively.

Table 2: Posterior means and 95% credible intervals.

Parameter	Posterior Mean	2.5% CI	97.5% CI
v	6.2851	6.088	6.482
r_b	0.7592	0.687	0.832
β	0.00619	0.00506	0.00732
δ	0.3562	0.307	0.406

6 Discussion and Conclusion

The results show strong agreement between least squares and Bayesian estimates. The Jacobian based covariance matrix yields small standard deviations and large t - values, supporting practical identifiability of all parameters under the chosen noise level. Bayesian inference provides more description of parameter uncertainty and reveals dependence between parameters. Posterior predictive simulations in Figure (5) confirm that the model predicts well within the calibration range and appropriately reflects increased uncertainty when extrapolating beyond the observed ages. The results highlight the value of combining classical least square method and Bayesian tools for parameter estimation in dynamical systems.

References

[1] Marko Laine. Mcmcstat: A matlab toolbox for bayesian inference. `https://mjlaine.github.io/mcmcstat/`, 2019. Accessed February 2026.

[2] The MathWorks, Inc. Matlab. Natick, Massachusetts, 2024. Version R2024b.